The use of postbiotic substances in rabbit breeding to sustain rabbit meat quality
Využitie postbiotických substancií v chove králikov pre udržanie kvality králičieho mäsa

Monika Pogány Simonová[1], Ľubica Chrastinová[†2], Eva Bino[1], Jana Ščerbová[1], Valentína Focková[1] Natália Zábolyová[1], Andrea Lauková[1]

[1]Centre of Biosciences of the Slovak Academy of Sciences. Institute of Animal Physiology, Košice, Slovakia
[2]National Agricultural and Food Centre, Nitra, Slovakia
simonova@saske.sk

Routine use of antibiotics (ATB) in agriculture – global concern to humans, animals and environmental health

- **alarming rise of ATB-resistant bacteria including methicillin-resistant staphylococci (MRS) in pets, food animals and in their products (meat, egg, milk)**
- **livestock – reservoir of resistant bacteria**
- **difficult treatment, zoonotic character, animal products contamination, risk to human health**
- **ecological burden (high migration of resistant microbiota into the environment)**

Rabbits – important sector of livestock production
- healthy meat, fur/wool, small sizes, rapid growth, high prolificacy, feed conversion and productive capacity
- sensitive animals - weaning period associated with stress, dietary changes, infections → negative impact on health, productivity, meat quality

Rabbit meat – lean, easily digestible meat
- high content of proteins, essential amino and fatty acids, vitamins B1, B6, B12, E, minerals (Ca, P, K, Fe, Se)
- low content of Na, cholesterol, purines

Natural substances - alternatives to synthetic ATB, reduction of MRS incidence

Postbiotics - metabolites of beneficial bacteria
Enterocins (Ent) – antimicrobial protein substances produced by *Enterococcus* spp.

AIM OF THE STUDY
spoilage/pathogenic environment simulation applying methicillin-resistant *Staphylococcus epidermidis* P3/Tr2a strain (MRSE), study of non-commercial EntA/P and Ent7420 to examine the effect of EntA/P and Ent7420 on the quality of rabbit meat.

Experimental scheme

rabbits - 35 days of age, weaned, meat line M91 and P91, both sexes

Ent7420, EntA/P
(50 µL/animal/day for 21 days; between 0-21 days in drinking water)

MRSE P3/Tr2a strain
(10^5 CFU/mL, 500 µL/animal/day between 0-21 days in drinking water)

wthdrawal of the substances (between 21-42 days)

- **Experimental group Ent7420 (n=22)**
- **Experimental group EntA/P (n=22)**
- **Experimental group MRSE (n=22)**
- **Experimental group Ent7420+MRSE (n=22)**
- **Experimental group EntA/P+MRSE (n=22)**
- **Control group (n=22)**

- **Meat samples of *Longissimus thoracis et lumborum* –**
- **pH24 -** Radelkis OP-109
- **total water, protein and fat content** (g/100g) - INFRATEC 1265
- **energy value -** [EC(kJ/100g) = 16.75 x Protein content + 37.68 x Fat content]

		MRSE	EntA/P	EntA/P+MRSE	Ent7420	Ent7420+MRSE	C
pH24	21	5.99 ± 0.05	5.92 ± 0.04	5.91 ± 0.04	6.00 ± 0.04	5.99 ± 0.04	5.99 ± 0.04
	42	5.99 ± 0.05	5.94 ± 0.05	5.96 ± 0.08	5.96 ± 0.06	6.02 ± 0.03	6.02 ± 0.02
Total water content (g/100 g)	21	74.97 ± 0.15	74.96 ± 0.37	74.40 ± 0.37	75.04 ± 0.18	74.77 ± 0.57	74.56 ± 0.22
	42	74.19 ± 0.79	73.82 ± 0.53	73.75 ± 0.32	74.78 ± 0.58	75.10 ± 0.53	74.49 ± 0.64
Total protein content (g/100 g)	21	22.84 ± 0.37	23.05 ± 0.19	23.07 ± 0.37	22.90 ± 0.17	22.78 ± 0.19	23.04 ± 0.40
	42	23.70 ± 0.24	23.62 ± 0.21	23.78 ± 0.46	23.54 ± 0.38	23.37 ± 0.52	23.67 ± 0.48
Total fat content (g/100 g)	21	1.16 ± 0.20	1.43 ± 0.20	1.38 ± 0.12	1.50 ± 0.18	1.33 ± 0.14	1.32 ± 0.12
	42	0.84 ± 0.11	1.12 ± 0.22	1.28 ± 0.13	1.24 ± 0.07	1.37 ± 0.26	0.95 ± 0.15
Energy value (kJ/100 g)	21	426.2 ± 11.1	439.8 ± 8.8	438.4 ± 2.2	440.1 ± 5.5	435.2 ± 9.2	434.3 ± 10.7
	42	437.8 ± 10.0	437.8 ± 10.0	446.3 ± 8.1	440.8 ± 7.1	442.8 ± 11.5	432.0 ± 13.3

MRSE application
- **no negative effect** on rabbit meat quality

EntA/P and Ent7420 application
- **no adverse effect** on meat physico-chemical parameters
- **improved** pH and the energy values

MRSE + EntA/P, Ent7420 application
- **optimized** fat content – **protective effect of Ents**

The great potential of EntA/P and Ent7420 as feed additives in the rabbit diet to sustain/improve meat quality during a possible staphylococcal infection.

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